

PROBABILITY THEORY

Session 2 - solutions

Problem 1

(b) $P(X > 1/2) = 1 - P(X \leq 1/2) = 1 - F(1/2) = 1 - \frac{1}{4} = 3/4$

(c) $P(2 < X \leq 4) = F(4) - F(2) = 1 - 11/12 = 1/12$

(d) $P(X < 3) = F(3) = 11/12$

(e) In order to answer 4.4(e) we are using the explanation from problem 4.5 (see below): $P(X = 1) = F(1+) - F(1-) = 2/3 - 1/2 = 1/6$

Problem 2

First, we determine the exact value of λ using the fact that $\int_{-\infty}^{\infty} f(x)dx = 1$:

$$\int_{-\infty}^{\infty} f(x)dx = \lambda \int_{-\infty}^{\infty} e^{-\frac{x}{100}} dx = \left[-100\lambda e^{-\frac{x}{100}} \right]_0^{\infty} = 100\lambda$$

Thus,

$$\lambda = \frac{1}{100}$$

Probability that a computer will function between 50 and 150 hours before breaking down:

$$\begin{aligned} P(50 \leq X \leq 150) &= \int_{50}^{150} \frac{1}{100} e^{-\frac{x}{100}} dx = \left(-e^{-\frac{x}{100}} \right) \Big|_{50}^{150} = \\ &e^{1/2} - e^{-3/2} \simeq 0.3834 \end{aligned}$$

Probability that a computer will function less than 100 hours:

$$\begin{aligned} P(X \leq 100) &= \int_0^{100} \frac{1}{100} e^{-\frac{x}{100}} dx = \left(-e^{-\frac{x}{100}} \right) \Big|_0^{100} = \\ &1 - e^{-1} \simeq 0.6321 \end{aligned}$$

Problem 3

from (a) we get $X = \{0, 1, 2, 3\}$

from (b) we get $P(X = 1) = P(X = 2)$

from (c) we get $P(X = 0) = P(X = 3)$

from (d) we get $P(X = 1) = \frac{1}{2}P(X = 0)$

Let $P(X = 1) = x$. Since the sum of all probabilities should give us 1, we have $x + x + 2x + 2x = 1$. Then $x = 1/6$ and we have the following pmf:

$P(X = 0) = 1/3$, $P(X = 1) = 1/6$, $P(X = 2) = 1/6$, $P(X = 3) = 1/6$.

Problem 4

The probability density function of X can be calculated as follows:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} x e^{-(x+y)} dy = x e^{-x} \int_0^{\infty} e^{-y} dy = \\ &= x e^{-x} \left[-e^{-y} \right]_0^{\infty} = x e^{-x} \end{aligned}$$

Thus,

$$f_X(x) = \begin{cases} x e^{-x}, & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

The probability density function of Y can be calculated as follows:

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = e^{-y} \int_0^{\infty} x e^{-x} dx = \left[e^{-x} (-x - 1) \right]_0^{\infty} e^{-y} = e^{-y}$$

Here are the details of integral calculations¹:

$$\int x e^{-x} dx = - \int x d e^{-x} = -x e^{-x} + \int e^{-x} dx = -x e^{-x} - e^{-x}$$

Thus,

$$f_Y(y) = \begin{cases} e^{-y}, & y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Noticing that $f(x, y) = f_X(x) f_Y(y)$, we can conclude that random variables X and Y are independent.

¹use formula $\int u dv = uv - \int v du$

Problem 5

If $f(x)$ is a pdf, then it should satisfy the following equality:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

Thus,

$$\begin{aligned} c \int_0^{\infty} e^{-2x} dx &= 1 \\ c \left(-\frac{1}{2} e^{-2x} \right) \Big|_0^{\infty} &= 1 \\ c \cdot \frac{1}{2} &= 1 \end{aligned}$$

Answer: $c = 1$

$$P(X > 2) = \int_2^{\infty} 2e^{-2x} dx = (-e^{-2x}) \Big|_2^{\infty} = e^{-4}$$

Problem 6

Lets define X as a number of retransmissions required to receive a message correctly. A r.v. X can take on the following values: 0; 1; 2; until infinity.

Lets find probability that a message was sent only once. It means that we had a single transmission, which was successful and there is no retransmissions. Thus, $P(X = 0) = 0.8$.

Event that a message is sent twice correspond to a case when the first time the message was lost and the second time it was received correctly. Thus, $P(X = 1) = 0.2 * 0.8$.

The message is sent three times in the case when it is lost twice and then it is received correctly. Thus, $P(X = 2) = 0.2^2 * 0.8$.

We can observe the pattern now. $X = k$ when the first k times the message is lost and during the last k th transmission it is received correctly. We can now write a formula for pmf for X , for $k \geq 1$:

$$P(X = k) = 0.2^k * 0.8$$

Problem 7

Let us consider a r.v. X_{head} representing a number of heads obtained in the measurements. X_{head} has a the following sample space:

$$X_{head} \in \{0, 1, 2, \dots, n\}$$

A r.v. X_{tail} represents a number of tails obtained in the measurements. It's sample space is:

$$X_{tail} \in \{0, 1, 2, \dots, n\}$$

We know that the sum of these r.v. is n : $X_{head} + X_{tail} = n$. We have to find a sample space for a r.v. $X = X_{head} - X_{tail}$.

$$X = X_{head} - X_{tail} = 2X_{head} - n$$

Therefore, the possible values of the r.v. X are:

$$X \in \{-n, -n + 2, -n + 4, \dots, n - 4, n - 2, n\}$$

Problem 8

a) Possible values for X and Y are $\{1, 2, 3, 4, 5, 6\}$. Assuming the dice are fair, all probabilities are equally likely: $P(X = k) = 1/6$ and $P(Y = j) = 1/6$, where k and j are numbers $\{1, 2, 3, 4, 5, 6\}$.

Since X and Y are independent random variables, we can write

$$P(X = k, Y = j) = P(X = k)P(Y = j) = 1/36$$

b)

$$P(X = 2, Y = 6) = P(X = 2)P(Y = 6) = 1/36$$

c) Since X and Y are independent, knowing the value of Y does not impact the probabilities for X , thus

$$\begin{aligned} P(X > 3|Y = 2) &= P(X > 3) = P(X = 4) + P(X = 5) + P(X = 6) \\ &= 1/6 + 1/6 + 1/6 = 1/2 \end{aligned}$$